

A
Mini Project Report
on
**AI POWERED VEHICLE DAMAGE ASSESSMENT AND
COST ESTIMATION FOR INSURANCE CLAIMS**

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Bachelor of Technology
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CERTIFICATE

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ABSTRACT

Nowadays, a lot of money is being wasted in the car insurance business due to leakage claims. Claims leakage /Underwriting leakage is characterized as the discrepancy between the actual payment of claims made and the sum that should have been paid if all of the industry's leading practices were applied. Visual examination and testing have been used to may these results. However, they impose delays in the processing of claims. The insurance industry is undergoing a digital transformation, and artificial intelligence (AI) is playing a pivotal role in streamlining various processes. One of the critical areas in the insurance sector is vehicle damage assessment and cost estimation for insurance claims. Traditional methods for assessing vehicle damage and determining repair costs are time-consuming, error-prone, and often lead to disputes between insurance companies and policyholders. To address these challenges, AI-powered solutions have emerged to enhance the accuracy and efficiency of vehicle damage assessment and cost estimation for insurance claims. Our mini-project focuses on the development of an AI-powered system that leverages computer vision, machine learning, and data analytics to assess vehicle damage and estimate repair costs. The aim of this project is to build a VGG16(CNN), computer vision model that can detect the area of damage on a car. The rationale for such a model is that it can be used by insurance companies for faster processing of claims if users can upload pics and the model can assess damage (be it dent scratch from and estimates the cost of damage). This model can also be used by lenders if they are underwriting a car loan especially for a used car.

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CHAPTER 1

INTRODUCTION

The "AI-Powered Vehicle Damage Assessment and Cost Estimation for Insurance Claims" project is a comprehensive solution aimed at transforming the insurance industry's claims processing by leveraging artificial intelligence (AI) and computer vision technologies. The project's primary objective is to streamline the assessment of vehicle damage and the estimation of repair costs, providing insurance companies with more accurate, efficient, and cost-effective ways to handle claims. Today, one of the first businesses to invest in innovation, cutting-edge technology, and artificial intelligence (AI) is the insurance industry. Car insurance companies spend millions of dollars each year due to evasion of insurance claims in today's society where the number of car accidents is on the rise.. In the insurance industry, Artificial intelligence (AI) based on machine learning and deep learning can assist with challenges including data analysis and processing, fraud detection, risk reduction, and claim automation. As a result, insurance companies have sought to reduce the time it takes to analyze damage and settle claims. However, developing current applications to solve these problems remains difficult, especially when using deep learning to assess car damage. Deep learning is an effective method for tackling complicated problems, but it necessitates more resources for model building, i.e., deep learning demands a large dataset and takes longer to compute. There are no publicly available datasets for automobile damaged photos because car damage assessment is a specialized topic.

1.1 Problem Statement

The car insurance industry faces significant financial losses due to claims leakage, arising from discrepancies between actual and expected claim payments. Traditional methods of visual examination and testing for vehicle damage assessment and cost estimation are time-consuming, error-prone, and contribute to delays in claims processing. To address these challenges, there is a critical need for a streamlined, efficient, and accurate solution leveraging the power of artificial intelligence (AI) in the form of computer vision and machine learning.

1.2 Project Objective and Scope

The primary objective of this mini-project is to design and implement an AI-powered system based on the VGG16 CNN architecture for accurate vehicle damage assessment and cost estimation. The scope encompasses the development of a computer vision model that can analyze images of damaged cars uploaded by users.

The model should reliably detect and classify areas of damage, such as dents or scratches, and provide an estimate of the repair costs. This solution aims to significantly reduce claims processing time for insurance companies, minimize disputes, and enhance overall operational efficiency. Moreover, the project extends its applicability to lenders involved in car loan underwriting, offering them a valuable tool to assess the condition and potential risks associated with used vehicles. The success of this project could lead to a transformative impact on the insurance and lending industries by leveraging AI for more accurate, efficient, and streamlined processes.

CHAPTER 2

LITERATURE SURVEY

The literature survey for the project "AI-Powered Vehicle Damage Assessment and Cost Estimation for Insurance Claims" reveals a growing body of research in the intersection of artificial intelligence, computer vision, and insurance claims processing. Existing studies emphasize the significance of leveraging AI algorithms, particularly machine learning models, for enhancing accuracy and efficiency in assessing vehicle damages and estimating associated costs. Several works highlight the successful application of computer vision techniques in automating the identification of diverse types of damages, including dents, scratches, and structural issues. Convolutional neural networks (CNNs) and other deep learning architectures to extract meaningful features from high-resolution images, contributing to the development of robust damage assessment models.

The literature underscores the need for objective and transparent insurance claims processes, emphasizing how AI technologies address the limitations of traditional human-centric assessments. AI systems to learn from historical data, improving over time and adapting to evolving patterns of vehicle damage. AI-powered solutions on reducing claims settlement time, ultimately enhancing customer satisfaction.

The aim is to develop an AI-powered system capable of detecting and assessing damaged areas on cars, using a VGG16-based convolutional neural network (CNN). This system would enable insurance companies to expedite claims processing by allowing users to upload images for automated damage assessment and cost estimation. Additionally, the model can be valuable for lenders involved in underwriting car loans, particularly for used cars, providing a comprehensive evaluation of the vehicle's condition.

CHAPTER 3

SYSTEM ANALYSIS

3.1 Existing System

The existing system for vehicle damage assessment and cost estimation in insurance claims predominantly relies on manual inspections conducted by human appraisers. This process is inherently time-consuming, subjective, and prone to inconsistencies. Human appraisers visually inspect damaged vehicles to determine the extent of damages and estimate repair costs based on their expertise and historical data. However, this approach may lead to variations in assessments, potential biases, and delays in claims processing.

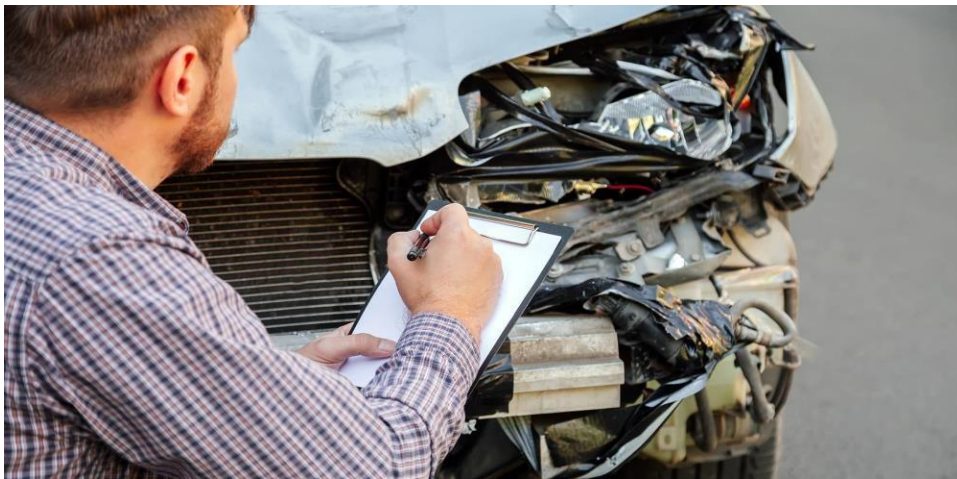


Figure 1: Manual Damage Assessment

The insurance industry faces several challenges related to vehicle damage assessment and cost estimation for insurance claims, including:

- **Manual and Time-Consuming Processes:** Traditional vehicle damage assessment relies heavily on manual inspection by human assessors, leading to time-consuming and inefficient claims processing.
- **Fraudulent Claims:** Insurance fraud, including exaggerated or fabricated damage claims, remains a significant problem, costing insurance companies substantial financial losses.
- **High Operational Costs:** Employing a large number of assessors and conducting on-site inspections can be expensive for insurance companies.

3.2 Proposed System

A real-time, open-source computer vision, utilizing Convolutional Neural Networks to autonomously detect car damage in images, Data analytics is integrated for decision insights and transparent documentation, enhancing the overall evaluation and cost estimation process for insurance claims, methods using the approach of NLP mechanism.

- **Automated Damage Assessment:** We propose the development of machine learning models capable of automatically and accurately assessing the severity and location of vehicle damage by analyzing images of the damaged vehicles. This automation eliminates the need for manual inspections, reducing human subjectivity and inconsistency in damage assessments.
- **Real-time Evaluation:** Our solution provides real-time damage assessment, allowing insurance companies to process claims rapidly. By streamlining this critical step, we aim to significantly reduce the time required for claims settlements, improving customer satisfaction and operational efficiency.
- **Cost Estimation:** To ensure precise repair cost estimation, our solution considers the identified damage severity and location and utilizes historical repair cost data. This data-driven approach minimizes the likelihood of inaccurate cost estimates and helps insurance companies allocate the appropriate funds for repairs.



Figure 2: Automated Damage Assessment

CHAPTER 4

SYSTEM SPECIFICATIONS

4.1 Hardware Requirements

- RAM – 4GB
- 1TB Hard Disk
- Any Intel Core i5, i7
- Camera

4.2 Software Requirements

- Python IDLE
- Anaconda
- Jupyter Notebook
- OS – Windows 7,8 or 10 (32 or 64 bit)

CHAPTER 5

TECHNOLOGIES USED

5.1 What is Deep Learning?

Deep learning is a branch of machine learning which is based on artificial neural networks. It is capable of learning complex patterns and relationships within data. In deep learning, we don't need to explicitly program everything. It has become increasingly popular in recent years due to the advances in processing power and the availability of large datasets. Because it is based on artificial neural networks (ANNs) also known as deep neural networks (DNNs). These neural networks are inspired by the structure and function of the human brain's biological neurons, and they are designed to learn from large amounts of data.

5.2 What is Deep Learning used for?

Deep learning is a machine learning technique that teaches computers to do what comes naturally to humans: In the project of "AI-powered Vehicle Damage Assessment and Cost Estimation for Insurance Claims," deep learning, and specifically deep neural networks, plays a pivotal role, contributing to the project's core capabilities. The utilization of deep learning techniques, such as Convolutional Neural Networks (CNNs), exemplified by the VGG16 architecture, facilitates the automatic recognition and analysis of vehicle damages from images. The deep learning model learns to generalize and accurately detect damage patterns. The integration of deep learning thus not only expedites the claims processing for insurance companies but also enhances the accuracy and consistency of vehicle damage assessment in comparison to traditional, manual methods.

5.3 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are a specialized class of deep learning models designed for image-related tasks. Their architecture is inspired by the human visual system, leveraging convolutional layers to automatically learn hierarchical representations of features within images. The key innovation of CNNs lies in their ability to recognize patterns in a spatial hierarchy, allowing them to identify complex visual structures. Each convolutional layer applies filters or kernels to extract specific features, such as edges, textures, and shapes, from the input image. Subsequent layers build upon these extracted features, enabling the network to progressively learn more abstract and complex representations.

The utilization of VGG16 in our project not only facilitates rapid and precise damage assessment but also aligns with industry-leading practices, ensuring a cutting-edge and reliable solution for insurance companies. As users upload images of damaged vehicles, the VGG16-based CNN swiftly processes the visual data, providing a comprehensive and automated evaluation of the damage. This integration of advanced neural network technology not only accelerates the claims processing for insurance companies but also establishes a foundation for transparent, consistent, and data-driven vehicle damage assessments, ultimately reshaping the future of insurance claims management.

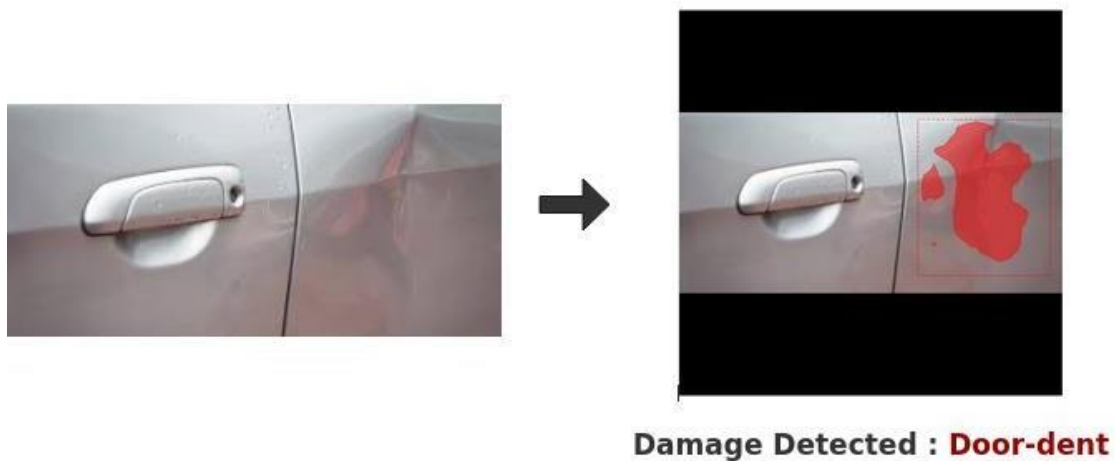


Figure 3: Car damage assessment using Deep Learning

5.4 Applications of CNN

- **Automated Image Recognition for Damage Assessment:** Convolutional Neural Networks (CNNs) play a pivotal role in automating image recognition for vehicle damage assessment. Trained on extensive datasets comprising diverse instances of car damages, CNNs, such as VGG16, excel at identifying patterns associated with various types of dents, scratches, and other damages. This application ensures a rapid and accurate assessment of the extent and location of damage, providing insurance companies with a powerful tool for claims processing.
- **Efficient Claims Processing:** The integration of CNNs like VGG16 facilitates efficient claims processing by automating the assessment of vehicle damages. As users upload images of damaged vehicles, the AI-powered system swiftly processes the visual data, significantly reducing the time and resources required for manual inspections. This application streamlines the claims processing workflow, expediting the overall insurance claims lifecycle.
- **Objective Damage Quantification:** CNNs contribute to the objective quantification of vehicle damage by eliminating subjective human biases. The VGG16 model, through its deep learning capabilities, learns to quantify damages based on visual cues and historical data. This objectivity enhances the consistency and reliability of damage assessments, minimizing disputes between insurance companies and policyholders.
- **Real-time Damage Visualization:** CNNs enable real-time damage visualization, allowing insurance professionals to observe and analyze the extent of damages promptly. The VGG16 model processes images in near real-time, providing a dynamic visual representation of the damages. This application enhances decision-making processes and ensures a comprehensive understanding of the vehicle's condition.
- **Enhanced Risk Evaluation for Lenders:** Beyond insurance claims, CNNs find applications in car loan underwriting, especially for used vehicles. Lenders can leverage the VGG16 model to assess the condition of a vehicle comprehensively. This contributes to enhanced risk evaluation, allowing lenders to make informed decisions when underwriting car loans, thereby minimizing potential financial risks associated with used vehicles.

CHAPTER 6

SYSTEM DESIGN

6.1 System Architecture

The different steps which will be carried out in order to create and use different classifiers and get most probable results. Out of the different results and accuracies produced by different Machine Learning models, we will utilize the best model with the best accuracy for Car damage detection and price prediction. System Architecture will give us an overview of the working of the system.

- **User Interface:** The user interface serves as the front end of the system, allowing users, including insurance claimants or lenders, to interact with the system. Users can upload images of damaged vehicles for assessment and receive the AI-generated damage reports and cost estimates.
- **Image Data Preprocessing:** Before inputting images into the neural network, a preprocessing module is employed to standardize and enhance the input data. This may include resizing, normalization, and other image processing techniques to ensure consistency in the input data.
- **Convolutional Neural Network (CNN):** The core of the system, the CNN, particularly utilizing the VGG16 architecture, is responsible for image analysis and damage recognition. Trained on a diverse dataset of vehicle damages, the CNN extracts features, detects patterns, and accurately identifies the location and extent of damages in the uploaded images.
- **Damage Classification and Localization:** Following the image analysis by the CNN, a module classifies the type of damage (e.g., dent, scratch) and localizes the areas affected on the vehicle. This information is crucial for generating detailed damage reports.
- **Cost Estimation Model:** A machine learning model, possibly based on regression techniques, is employed to estimate the repair costs associated with the identified damages. This model learns from historical data and market trends to provide accurate cost predictions.
- **Database Management System:** The system integrates a robust database management system (DBMS) to store and manage a vast dataset of vehicle images, historical claims data, and repair cost information. This database is crucial for training and updating the machine learning models.

- **Integration with External Data Sources:** The system may integrate with external data sources, such as industry databases, auto repair cost databases, and market trends, to enhance the accuracy of damage assessment and cost estimation. Real-time updates from these sources contribute to the system's reliability.
- **Application Programming Interface (API):** An API facilitates communication between different system modules and external applications. It allows for seamless data flow and integration with other tools or platforms that insurance companies or lenders might use.
- **Reporting and Visualization Module:** The reporting module generates comprehensive reports detailing the identified damages, their classification, and the estimated repair costs. Visualization tools may be incorporated to provide graphical representations of the damages for better understanding.
- **Security and Compliance Layer:** Ensuring the security and compliance of sensitive data is paramount. This layer implements security protocols, encryption, and compliance measures to protect user data and adhere to industry regulations.

This architecture ensures a cohesive and efficient system capable of transforming the vehicle damage assessment and cost estimation process in insurance claims, providing a user-friendly interface and accurate, data-driven insights for insurance companies and lenders alike.

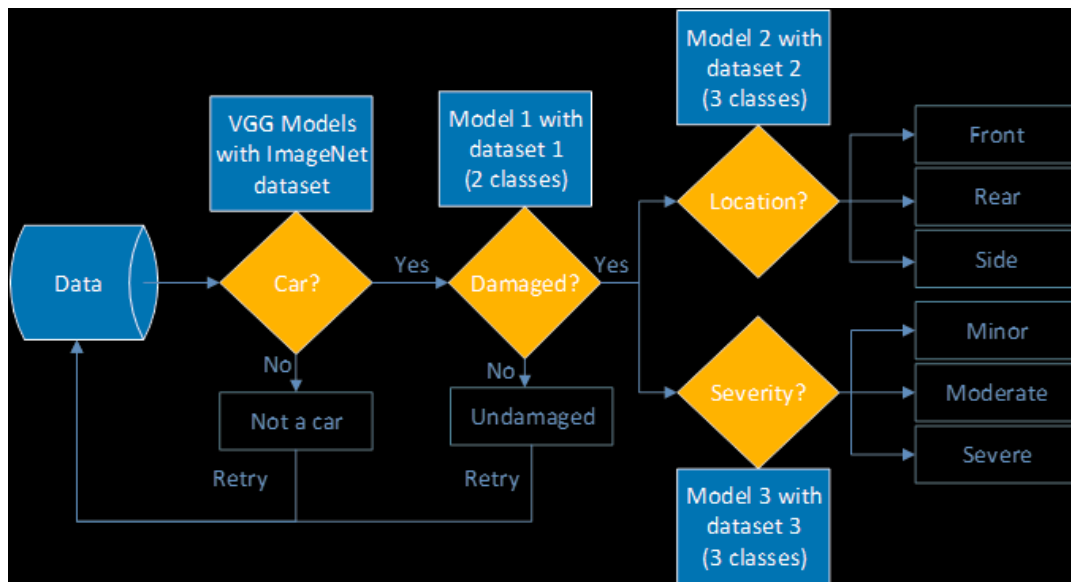


Figure 4: Flow chart of developing car damage assessment pipelines

- **Web App**

Here users can upload images of the damaged car which will be processed by our algorithm using the GUI which is created using HTML, CSS and integrated by flask.

- **Module-1**

In this module, VGG-16 is used to detect whether the uploaded image is a car or not.

- **Module-2**

Here, car damage is detected using VGG-16. The model that gives us the most accuracy is considered..

- **Module-3**

The side of the vehicle where the damaged is incurred i.e. In the rear part, front part or side part is identified using this module.

- **Module-4**

The damage level is further decided in this module. The level of damage is decided by data visualization.

- **Module-5**

Depending on the damage, the location of the damage and the model of the car, the cost of repairing the damage will be predicted.

CHAPTER 7

IMPLEMENTATION

The analysis of car damage model accepts an input image from an user in JPEG format and processes it across different stages, the project is divided into 2 phases, to identify the damaged car that is, if the car is damaged or not, and then to find the location of damage like front, rear or back. The analysis can be achieved using steps that include image classification and object detection. In image classification, it processes and classify the image provided by the user into either damaged car or a whole car i.e., undamaged or completely fine. Whereas, object detection and image localization come into picture to help us identify the location of the damage. It can predict the location along with the class for each object using Object Detection.

7.1 DATA ANNOTATION PROCESS

In the annotation process for fine-grained tasks like instance segmentation, ensuring consistency across human annotators is a significant challenge. Consistency is crucial for reliable deep-learning models but is both expensive and time-consuming to achieve. To enhance annotation quality, an iterative approach was employed. Initially, small batches of 20 images were annotated by three human annotators, aiming to refine instructional materials for consistent labeling. Inter-annotator consistency was assessed using the intersection over union (IoU), with low IoU indicating inconsistent annotations, prompting instructional material revisions. Through iterations, a labeling approach ensuring consistency emerged. Large batches were then annotated, and results were reviewed.

7.2 DATASET CREATION

The datasets used for each sub-task of the overall damage severity evaluation problem

7.2.1 CAR DETECTOR DATASET

For training the car detector we used the COCO dataset. For evaluation, we used only images containing relevant vehicle objects classes i.e., car, truck and bus. The distribution of images within each class.

7.2.2 INTERIOR-EXTERIOR DATASET

For the Interior-Exterior dataset, we collected around 3500 exterior images and 1600 interior images from



FIGURE 2. Interior samples of the Interior/Exterior Dataset.



FIGURE 3. Samples of the car angle classifier's dataset. Front view (Left) and Rear Left view (Right).

Figure 5: Interior/Exterior Car angle classifier's dataset

7.2.3 CAR ANGLE DATASET

We used a set of public car images retrieved from online sources for the car angle dataset. Images were manually labelled with one of 8 angle labels. The angle labels used and the distribution of samples in the train, validation and test sets. The dataset was augmented by applying horizontal flipping to every image. For non frontal/rear images this gives the view from the opposite side of the car e.g., flipping a front-right view creates a new image with the front-left view. This assumes a symmetrical exterior appearance, which is true in most non-damage cases. The dataset had a total of 3577 images and was split into 82% for training, 11% for validation and 7% for testing.

7.2.4 PARTS AND DAMAGE DATASET

The parts and damage datasets are described together as both datasets are similar. The parts dataset contained 8005 images.



Figure 6: Annotated car parts used in the car parts dataset

The dataset comprises 9,226 images, with each image annotated for part classification and damage labeling. For part classification, 24 classes were assigned to various car parts, while damage was categorized into five types. Segmentation masks were created for all visible damage and part boundaries. Annotations and labels, including segmentation masks, were generated using the AWS GroundTruth annotator. The same tool was employed for both parts and damage annotation. The distribution of damage samples

7.2.5 CLAIMS DATASET

The purpose of our overall pipeline is to predict the damage cost estimate by combining structured information provided by the claimant with knowledge about the extent of damage measured using the computer-vision pipeline. To train a model to predict the cost estimate given this information, we use a dataset of around 18,000 claims containing over 200,000 images for training and a dataset of 2,600 claims containing 30,000 images for validation.

7.2.6 AIRBAG DATASET

The airbag dataset contains 2560 airbag and 2381 non-airbag car images. These images were collected from a combination of public and private sources. Example interior and exterior airbag deployment images.



Figure 7: Airbag Deployment of images

7.2.7 PARTS AND DAMAGE DETECTION MODELS

The damage detection task imposes some unique challenges compared with object detection or parts segmentation. For instance, the damage detection model has to deal with detecting damaged areas with arbitrary shapes. We classify damages into five classes; Damage, Broken Light, Broken Window, Dent and Missing Part. Figure illustrates each type of damage, with the corresponding annotation. The damage class is the most severe. It is used when a panel is destroyed or when over half the panel is severely crumpled.

The damage class is the most severe. It is used when a panel is destroyed or when over half the panel is severely crumpled. Damaged vehicles can be very deformed, making the task of damage detection more challenging. We, therefore, used deformable convolutions as kernels in the damage detection network backbone, which improves performance. For the parts detection task, a total of 24 different car parts are detected. Both the parts detection and damage detection models.

We also take advantage of the similarities between the parts and damage detection problems. We treat both as segmentation problems where the objective is to predict a mask corresponding to the shape/area of the part or damage. Specifically, an instance segmentation approach is useful here. We use instances of parts and instances of damage to measure the extent of damage on each panel of the vehicle.



Figure 8: Example annotation from car damage dataset

CHAPTER 8

RESULTS

User Interface has been developed. Also the first module which detects the car has been integrated using flask.

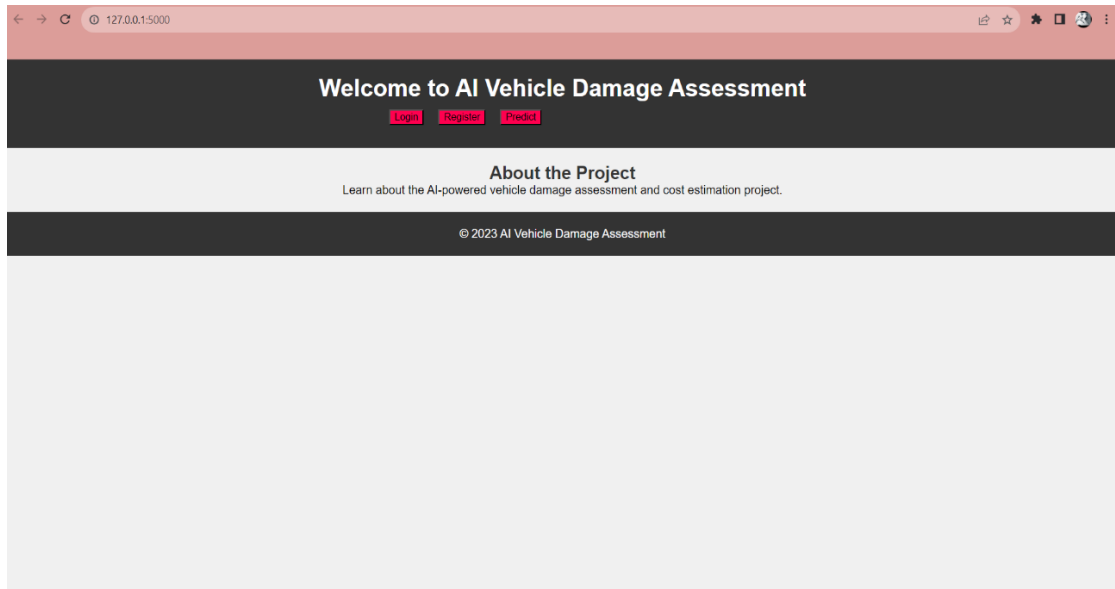


Figure 9: Home Page

Home page provides basic interface with the option showing Register Option.

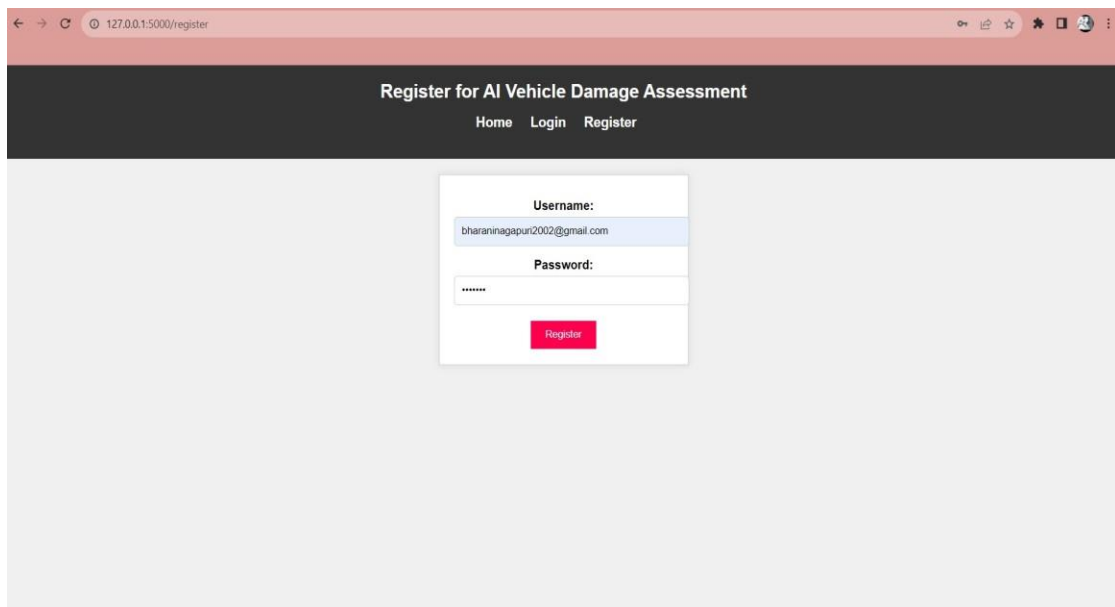


Figure 10: Register Page

Register page provides basic interface with the option showing Login with Predict option.

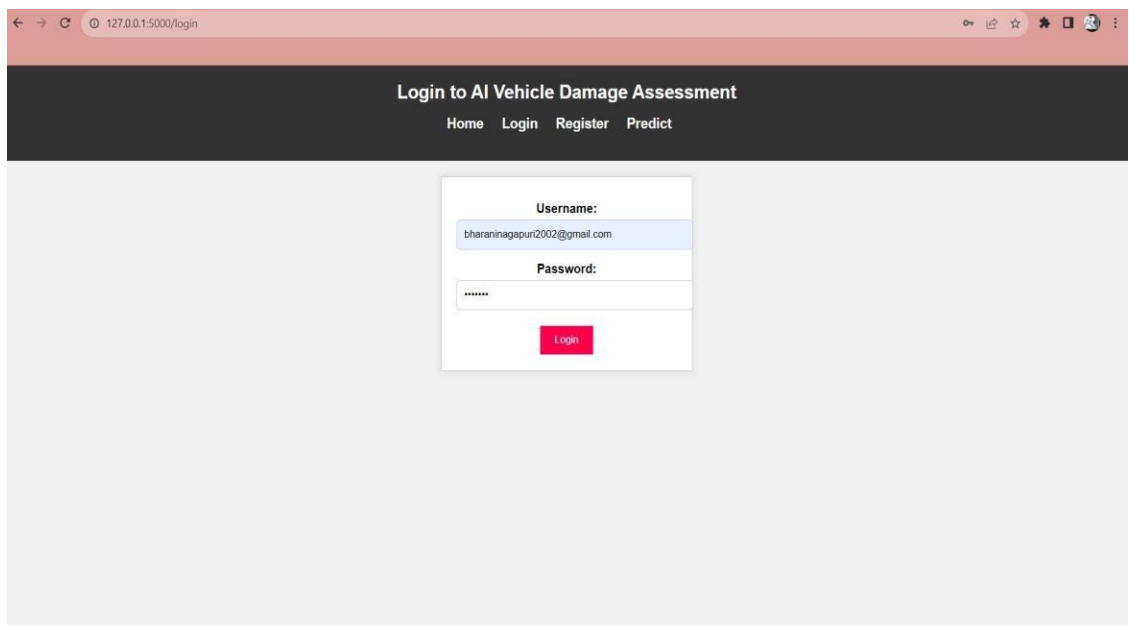


Figure 11: Login Page

Upload Vehicle Image.

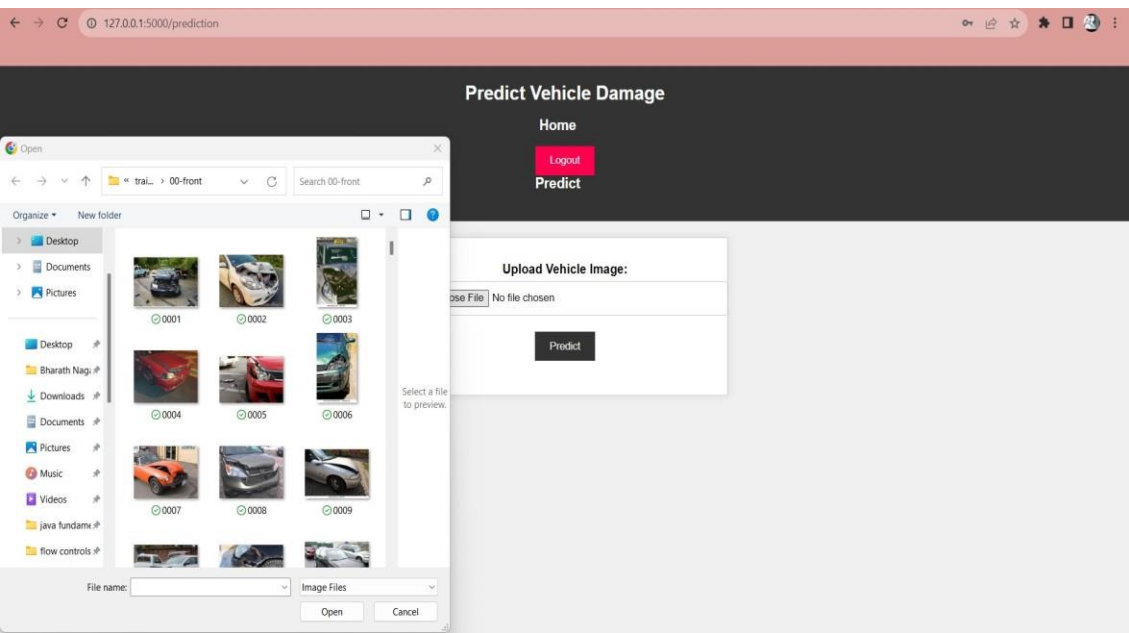


Figure 12: Choose car image

Predicting the output of damaged vehicle.

The screenshot shows a web browser window with the address bar displaying '127.0.0.1:5000/predict'. The page has a dark header with the title 'Predict Vehicle Damage' and navigation links for 'Home', 'Logout', and 'Predict'. The main content area is light gray and contains a white box with the heading 'Upload Vehicle Image:'. Inside this box, there is a file selection interface with a 'Choose File' button and the text 'No file chosen'. Below this is a 'Predict' button. At the bottom of the white box, the text 'The predicted output is 7000 - 9000 INR' is displayed.

Figure 13: Cost estimation of damaged vehicle

CHAPTER 9

CONCLUSION

In conclusion, this project "AI-Powered Vehicle Damage Assessment and Cost Estimation for Insurance Claims" offers a transformative solution to the challenges faced by the insurance and automotive industries. By leveraging artificial intelligence, it enables faster and more accurate assessment of vehicle damage, expedites claims processing, and enhances the overall customer experience. The system's ability to provide fair and data-driven cost estimations not only benefits policyholders but also contributes to fraud detection and risk assessment for insurance companies. With applications ranging from reduced manual work for claims adjusters to data insights for auto manufacturers, this project has the potential to significantly improve the efficiency and effectiveness of the insurance claims process. It represents a forward-looking approach to modernizing the insurance industry and optimizing the evaluation of vehicle damage, ultimately leading to increased customer satisfaction and operational efficiency.

CHAPTER 10

FUTURE SCOPE

The future scope of AI-powered vehicle damage assessment and cost estimation for insurance claims holds immense potential for further advancements and broader applications. As technology continues to evolve, several avenues present exciting opportunities for enhancement and expansion. Firstly, the incorporation of more advanced deep learning architectures beyond VGG16, such as state-of-the-art models like ResNet or efficient networks, can potentially improve the accuracy and efficiency of damage assessment. Additionally, the integration of advanced natural language processing (NLP) algorithms could enable the system to analyze textual data, such as repair invoices or claim descriptions, providing a more comprehensive understanding of the overall claims process. Furthermore, the utilization of real-time data from connected vehicles and IoT devices could revolutionize the industry by offering continuous monitoring and assessment of vehicle conditions.

CHAPTER 11

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